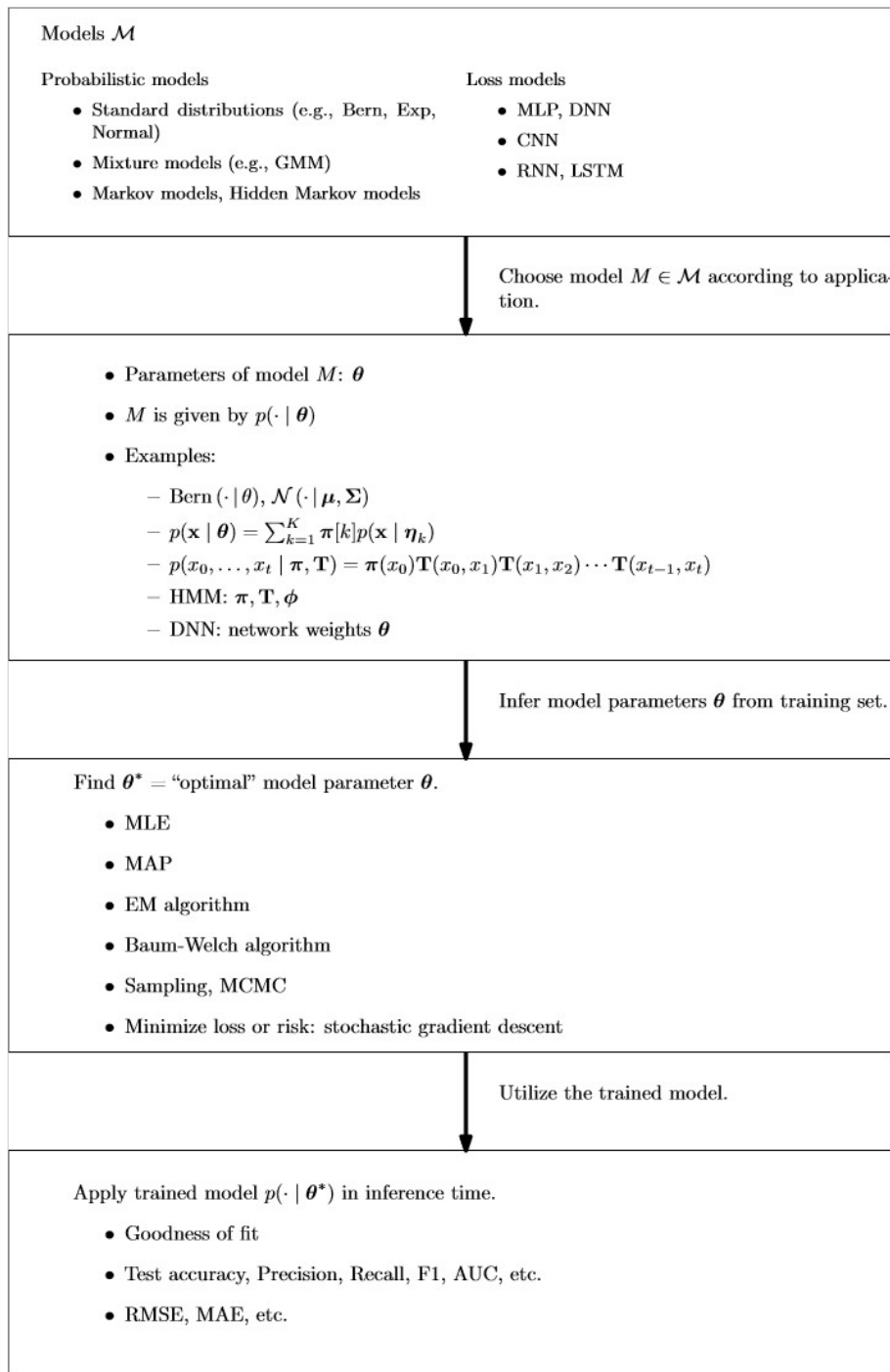




_overview



1)

$k=0$ $k=1$
 $r_k = \# \text{ of vulnerable for ped./cyclist}$

$\text{likelihood} \propto \theta_k^{r_k} (1 - \theta_k)^{n_k - r_k}$

$\uparrow$

$\log(\quad)$

$r_k \log \theta_k + (n_k - r_k) \log(1 - \theta_k)$

$\frac{\partial}{\partial \theta_k}$

$\frac{r_k}{n} - \frac{n_k - r_k}{1 - n} = 0$

$n_0 = 100$ $n_1 = 50$
 $r_0 = 20$ $r_1 = 5$

$$\frac{\partial}{\partial \theta_k} \left(\frac{r_k}{\theta_k} - \frac{n_k - r_k}{1 - \theta_k} \right) = 0$$

$$\text{MLE} \quad \hat{\theta}_{k, \text{MLE}} = \frac{r_k}{n_k} \quad \hat{\theta}_0 = 0.2, \quad \hat{\theta}_1 = 0.1$$

$$\text{prior} : \propto \theta^{a-1} (1-\theta)^{b-1} = \theta (1-\theta)^2$$

$$\log \text{prior} : \log \theta_k + 2 \log (1 - \theta_k)$$

$$\log \text{likelihood} + \log \text{prior}$$

$$= r_k \log \theta_k + (n_k - r_k) \log (1 - \theta_k) + \log \theta_k + 2 \log (1 - \theta_k)$$

$$= (r_k + 1) \log \theta_k + (n_k - r_k + 2) \log (1 - \theta_k)$$

$$\frac{\partial}{\partial \theta_k} : \frac{r_k + 1}{\theta_k} - \frac{n_k - r_k + 2}{1 - \theta_k} = 0$$

$$\text{MAP} \quad \hat{\theta}_{k, \text{MAP}} = \frac{r_k + 1}{n_k + 3} \quad \hat{\theta}_{0, \text{MAP}} = \frac{21}{103}$$

$$\hat{\theta}_{1, \text{MAP}} = \frac{6}{53}$$

pedestrian $\text{Bern}(\theta_0)$

cyclist $\text{Bern}(\theta_1)$

$$p(x|\theta) = \frac{4}{5} \text{Bern}(x|\theta_0) + \frac{1}{5} \text{Bern}(x|\theta_1) \quad \theta = (\theta_0, \theta_1)$$

$$\log p(x_1, \dots, x_{100} | \theta) = \sum_{i=1}^{100} \log p(x_i | \theta)$$

$$= \sum_{i=1}^{100} \log \left(\frac{4}{5} \theta_0^{x_i} (1 - \theta_0)^{1-x_i} + \frac{1}{5} \theta_1^{x_i} (1 - \theta_1)^{1-x_i} \right) \quad \checkmark$$

EM algo.

$$\downarrow \quad N(x|y, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-y)^2}{2}}$$

in arg.

$$N(x|y, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}}$$

$$\begin{aligned}
 2) \quad A(x, y) &= \min \left(1, \frac{\tilde{\pi}(y) q(y, x)}{\tilde{\pi}(x) q(x, y)} \right) \\
 &= \min \left(1, \frac{e^{-|y|} N(x|y, 1)}{e^{-|x|} N(y|x, 1)} \right) \quad \begin{matrix} x=1 \\ y=2 \end{matrix} \\
 &= \frac{e^{-2} \cdot e^{-\frac{(1-2)^2}{2}}}{e^{-1} \cdot e^{-\frac{(2-1)^2}{2}}} \\
 &= \boxed{e^{-1}} \\
 &= e^{-1} \simeq 0.3679
 \end{aligned}$$

$$\begin{array}{ccccccc}
 z_1 & \rightarrow & z_2 & \rightarrow & \dots & \rightarrow & z_i & \rightarrow & \dots \\
 \downarrow & & \downarrow & & & & \downarrow & & \\
 x_1 & & x_2 & & & & x_i & &
 \end{array} \rightarrow \begin{aligned}
 &P(x_i = D | z_i = D) = \alpha \\
 &P(x_i = U | z_i = D) = 1 - \alpha \\
 &P(x_i = D | z_i = U) = \beta \\
 &P(x_i = U | z_i = U) = 1 - \beta
 \end{aligned}$$

$$z_{i-1} \rightarrow z_i \rightarrow z_{i+1}$$

$$p(z_i | z_{i-1}, z_{i+1}) \propto \gamma^{1_{\{z_i = z_{i-1}\}} + 1_{\{z_i = z_{i+1}\}}}$$

$$D \rightarrow D \rightarrow D \quad \gamma^{1+1} = \gamma^2$$

$$D \rightarrow D \rightarrow U \quad \gamma^{1+0} = \gamma$$

$$\begin{aligned}
 (*) \quad & \left\{ \begin{aligned}
 p(z_i = D | z_{-i}, x) &\propto \underbrace{p(z_i = D | z_{-i})}_{\substack{\uparrow \\ (z_{i-1}, z_{i+1})}} \underbrace{p(x_i | z_i = D)}_{\substack{\uparrow \\ \{x_i = D\}}} \\
 &\propto \gamma^{1_{\{z_{i-1} = D\}} + 1_{\{z_{i+1} = D\}}} \cdot \alpha^{1_{\{x_i = D\}}} (1-\alpha)^{1_{\{x_i = U\}}} \\
 p(z_i = U | z_{-i}, x) &\propto \gamma^{1_{\{z_{i-1} = U\}} + 1_{\{z_{i+1} = U\}}} \beta^{1_{\{x_i = D\}}} (1-\beta)^{1_{\{x_i = U\}}}
 \end{aligned} \right.
 \end{aligned}$$

$$(*) \left\{ p(z_i = u | z_{-i}, x) \propto \gamma^{\mathbb{1}\{z_{i-1} = u\}} + \mathbb{1}\{z_{i+1} = u\} \beta^{\mathbb{1}\{x_i = D\}} (1-\beta)^{\mathbb{1}\{x_i = u\}}$$

- Initialize $(z_1^{(0)}, z_2^{(0)}, \dots, z_n^{(0)})$



- Iteration k :

- sample $z_1^{(k)} \sim p(z_1 | z_2^{(k-1)}, z_n^{(k-1)}, x)$

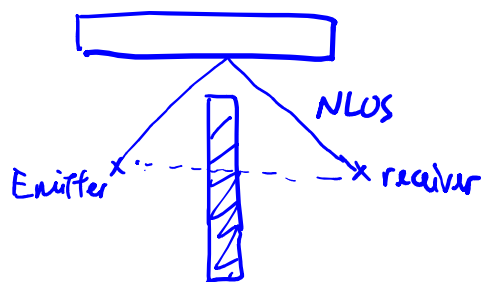
- sample $z_2^{(k)} \sim p(z_2 | z_1^{(k)}, z_3^{(k-1)}, x)$

- sample $z_3^{(k)} \sim p(z_3 | z_2^{(k)}, z_4^{(k-1)}, x)$

...

- sample $z_n^{(k)} \sim p(z_n | z_{n-1}^{(k)}, z_1^{(k)}, x)$

- Burnin



$$\begin{array}{c} \text{NLOS} \quad \text{LOS} \\ \downarrow \quad \downarrow \\ r = \underbrace{d} + \underbrace{\varepsilon} \geq d \\ \varepsilon \geq 0 \end{array}$$

$$\varepsilon \sim \text{Exp}(\lambda)$$

r	d	$\varepsilon = r - d$
3.4	3.0	0.4
2.8	2.5	0.3
4.5	4.3	0.2
2.4	2.3	0.1

$$(r, d) \rightarrow \varepsilon = r - d$$

"complete" data

$$\frac{\partial}{\partial \lambda} : \frac{4}{\lambda} - \sum_{i=1}^4 \varepsilon_i = 0$$

$$\hat{\lambda}_{MLE} = \frac{4}{\sum_{i=1}^4 \varepsilon_i} = \frac{4}{0.4 + 0.3 + 0.2 + 0.1}$$

$$= 4$$

$$\dots \cap \dots \sim n^{-\frac{1}{\alpha}}$$

$$p_{\text{prior}} = p(\lambda) \propto \lambda e^{-\frac{\lambda}{3}}$$

$$4 \log \lambda - \lambda \sum_{i=1}^4 \varepsilon_i + \log(\lambda e^{-\frac{\lambda}{3}}) + \text{const.}$$

$$= 5 \log \lambda - \lambda \left(\sum_i \varepsilon_i + \frac{1}{3} \right)$$

$$\frac{\partial}{\partial \lambda} : \quad \frac{5}{\lambda} - \left(\sum_i \varepsilon_i + \frac{1}{3} \right) = 0$$

$$\lambda = \frac{5}{\sum_i \varepsilon_i + \frac{1}{3}} = \frac{5}{1 + \frac{1}{3}} = \frac{15}{4}$$

$$r = d + \varepsilon$$

observe r $\rightarrow$ "incomplete"
 m th iteration

E step: conditional expectation of $\log p(r, d | \lambda)$

$$\text{w.r.t. } \underbrace{p(\underbrace{r, d}_{\lambda}, \lambda | \lambda^{(m-1)}, r)}_{\lambda} = \underbrace{p(d | \lambda^{(m-1)}, r)}_{=1} \cancel{p(r | \lambda^{(m-1)}, r)}$$